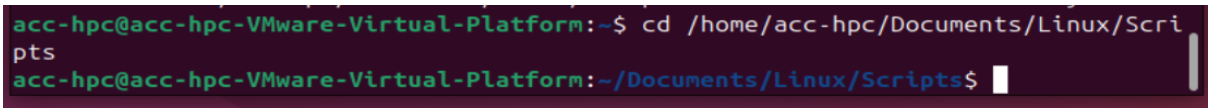
Name- Kaveri Avinash Jarhad

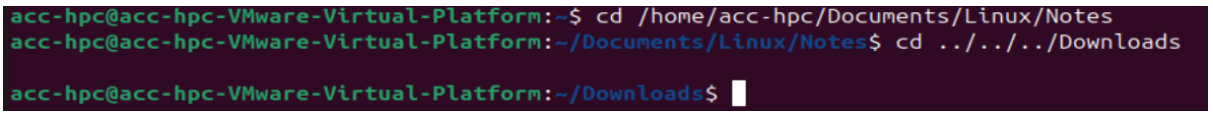
Q1. Navigate to the Scripts directory using an absolute path.



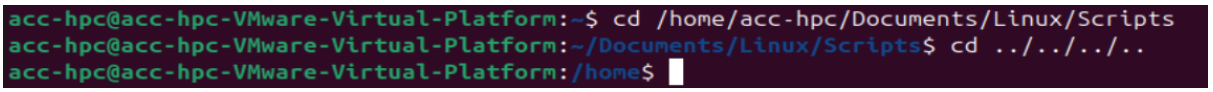
Q2. You are in /home/. Navigate to the Projects directory using a relative path.



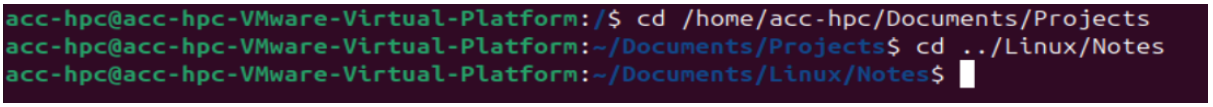
Q3. Your current directory is:
/home//Documents/Linux/Notes
Navigate to the Downloads directory using a relative path only.



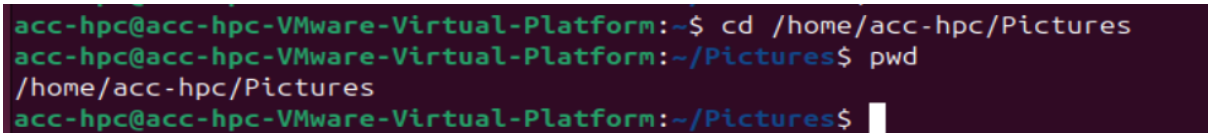
Q4. Your current directory is:
/home//Documents/Linux/Scripts
Go back to your home directory using the shortest possible command.



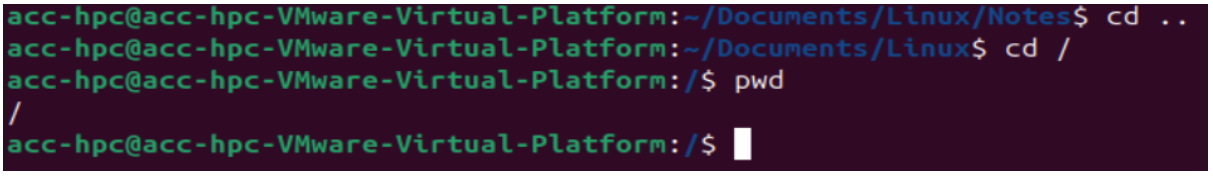
Q5. Your current directory is: /home//Documents/Projects Navigate to the Notes directory using a relative path.



Q6. Display your current working directory after navigating to the Pictures directory.
Expected Commands cd /home//Pictures pwd



Q7. You are in: /home//Documents/Linux Move to the root directory and verify your location.



Q8. Without changing your current directory, list the contents of the Downloads directory using an absolute path. (Current directory:) /home//Documents/Linux

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:/$ cd /home/acc-hpc/Documents/Linux
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$ ls -R /home/acc-hpc/Downloads
/home/acc-hpc/Downloads:
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$
```

Q9. Your current directory is: /home//Documents/Linux/Notes Navigate to the Music directory using a relative path.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux/Notes$ cd ../../../Music
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Music$
```

Q10. Starting from: /home/ Complete the following sequence:

1. Go to Documents/Linux
2. Move to its parent directory.
3. Enter Projects.
4. Return directly to your home directory.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$ cd /home/acc-hpc/Documents/Linux/Notes
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux/Notes$ cd ../../Music
bash: cd: ../../Music: No such file or directory
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux/Notes$ cd ../../../Music
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Music$ cd ../..
acc-hpc@acc-hpc-VMware-Virtual-Platform:/home$ cd /acc-hpc/Documents/Linux
bash: cd: /acc-hpc/Documents/Linux: No such file or directory
acc-hpc@acc-hpc-VMware-Virtual-Platform:/home$ cd /home/acc-hpc/Documents/Linux
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$ cd ..
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents$ cd Projects
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Projects$ cd ../../..
acc-hpc@acc-hpc-VMware-Virtual-Platform:/home$
```

Q11. Find and Display Matching Content Assume you are in: /home/student A file named employees.txt is present in the Documents directory.

Task: 1. Navigate to the Documents directory using a relative path

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$ cd Documents
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents$
```

. 2. Display the contents of employees.txt.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents$ cat employees.txt
101 Amit IT Pune 55000
102 Sneha HR Mumbai 48000
103 Rahul Finance Pune 62000
104 Priya IT Bangalore 58000
105 Karan Sales Delhi 45000
106 Neha IT Pune 67000
107 Ankit HR Pune 52000
108 Pooja Finance Mumbai 61000
109 Rohan Sales Pune 49000
110 Meera IT Hyderabad 70000
111 Suresh HR Bangalore 53000
```

3. Find all employees working in the IT department.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents$ grep -n "IT" employees.txt
1:101 Amit IT Pune 55000
4:104 Priya IT Bangalore 58000
6:106 Neha IT Pune 67000
10:110 Meera IT Hyderabad 70000
```

Q12. Environment Variables and Navigation You are currently in: /home/student/Documents/Linux Task

: 1. Display the value of the HOME environment variable

. 2. Navigate to your home directory using the environment variable (not cd ~).

3. List all files, including hidden files, in long listing format.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$ echo $HOME
/home/acc-hpc
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$ cd  $HOME
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$ ls -la
total 80
drwxr-x--- 15 acc-hpc acc-hpc 4096 Jul 27 18:51 .
drwxr-xr-x  3 root    root    4096 Jul 22 23:28 ..
-rw-------  1 acc-hpc acc-hpc 2365 Jul 27 20:14 .bash_history
-rw-r--r--  1 acc-hpc acc-hpc  220 Mar 31  2024 .bash_logout
-rw-r--r--  1 acc-hpc acc-hpc 3771 Mar 31  2024 .bashrc
drwx------  8 acc-hpc acc-hpc 4096 Jul 27 15:31 .cache
drwx------ 14 acc-hpc acc-hpc 4096 Jul 27 20:15 .config
drwxr-xr-x  5 acc-hpc acc-hpc 4096 Jul 27 18:50 Desktop
drwxr-xr-x  4 acc-hpc acc-hpc 4096 Jul 27 20:15 Documents
drwxr-xr-x  2 acc-hpc acc-hpc 4096 Jul 24 16:08 Downloads
-rw-------  1 acc-hpc acc-hpc   20 Jul 24 21:15 .lesshst
drwx------  4 acc-hpc acc-hpc 4096 Jul 24 16:08 .local
drwxr-xr-x  2 acc-hpc acc-hpc 4096 Jul 24 16:08 Music
drwxr-xr-x  2 acc-hpc acc-hpc 4096 Jul 24 16:08 Pictures
-rw-r--r--  1 acc-hpc acc-hpc  807 Mar 31  2024 .profile
drwxr-xr-x  2 acc-hpc acc-hpc 4096 Jul 24 16:08 Public
drwx------  5 acc-hpc acc-hpc 4096 Jul 24 21:54 snap
drwx------  2 acc-hpc acc-hpc 4096 Jul 25 00:12 .ssh
-rw-r--r--  1 acc-hpc acc-hpc    0 Jul 27 15:09 .sudo_as_admin_successful
drwxr-xr-x  2 acc-hpc acc-hpc 4096 Jul 24 16:08 Templates
drwxr-xr-x  2 acc-hpc acc-hpc 4096 Jul 24 16:08 Videos
```

Q13. Search Configuration Files You are in: /home/student A file named config.txt is located in: /home/student/Documents/Linux Task:

1. Display the contents of the Linux directory.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$ ls -R
.:
config.txt  Notes  Scripts

./Notes:

./Scripts:
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$
```

2. Search for the word server inside config.txt.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$ grep "server" config.txt
server
server
```

3. Display line numbers with the matching lines.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~/Documents/Linux$ grep -n "server" config.txt
3:server
5:server
```

Q14. Combine Navigation, Listing, and Searching Current directory: /home/student

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$ cd /home/acc-hpc | ls -la | find . -name "employees.txt"
./.local/share/Trash/files/employees.txt
./Documents/employees.txt
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$
```

Q15. Environment Variables, cat, grep, and ls You are logged in as user student.

Task: 1. Display all environment variables

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$ env
SHELL=/bin/bash
SESSION_MANAGER=local/acc-hpc-VMware-Virtual-Platform:@/tmp/.ICE-unix/2429,unix/acc-hpc-VMware-Virtual-Pl
E-unix/2429
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
MEMORY_PRESSURE_WRITE=c29tZSAyMDAwMDAgMjAwMDAwMAA=
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/acc-hpc
LOGNAME=acc-hpc
XDG_SESSION_DESKTOP=ubuntu
XDG_SESSION_TYPE=wayland
```

2. Find the environment variable related to the user's home directory.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$ env | grep USER
USERNAME=acc-hpc
USER=acc-hpc
```

3. List the contents of your home directory.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$ env | grep HOME
HOME=/home/acc-hpc
```

4. Display the contents of .bashrc.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$ cat .bashrc
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples

# If not running interactively, don't do anything
case $- in
    *i*) ;;
      *) return;;
esac

# don't put duplicate lines or lines starting with space in the history.
# See bash(1) for more options
HISTCONTROL=ignoreboth
```

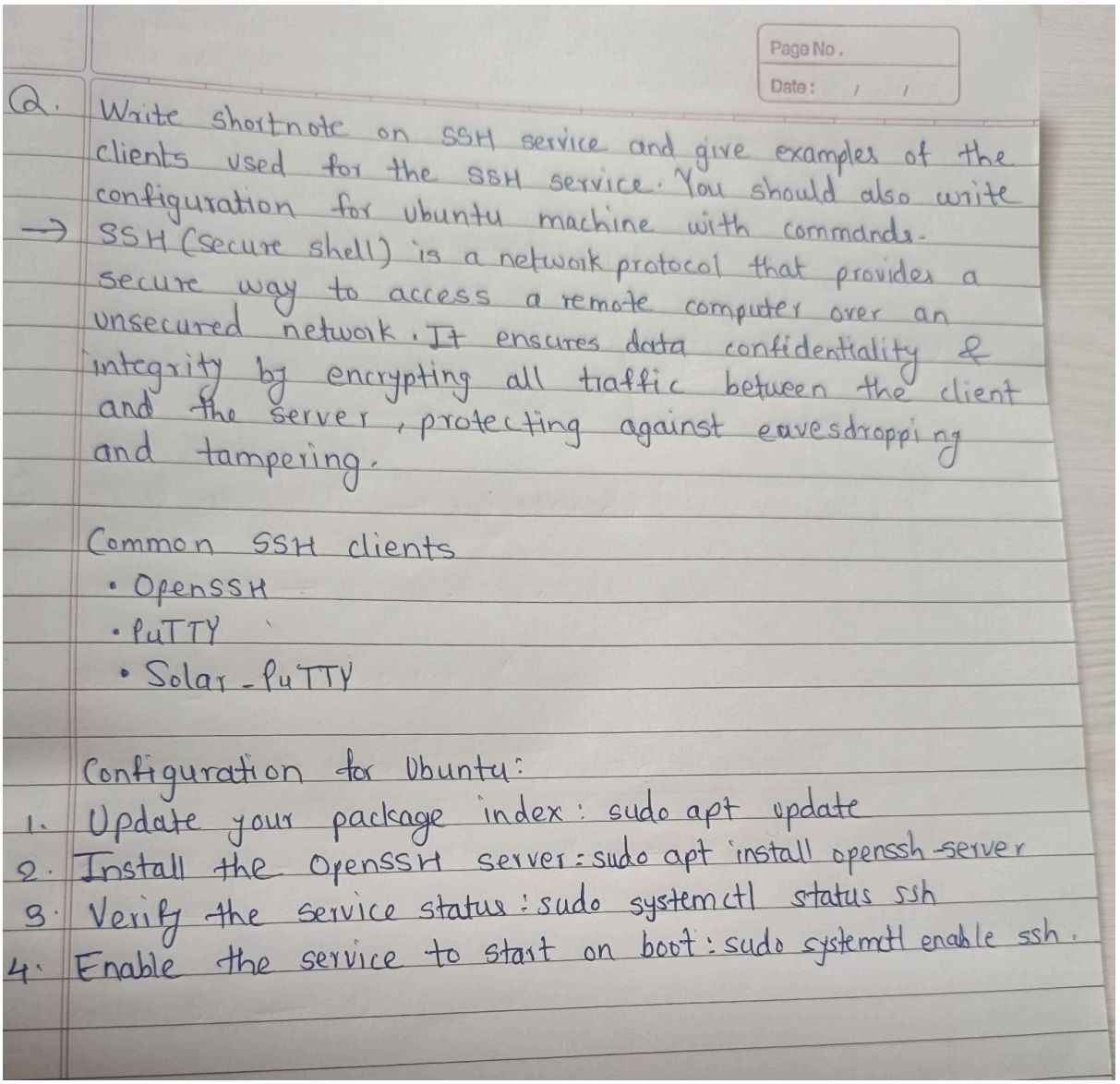
Q16) Print the output of last 9th command executed.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$ !-9
echo $HOME
/home/acc-hpc
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$
```

Q17) Print the output of last command executed.

```
acc-hpc@acc-hpc-VMware-Virtual-Platform:~$ !!
echo $HOME
/home/acc-hpc
```

Q18) Write shortnote on SSH service and give examples of the clients used for the SSH service. You should also write configuration for ubuntu machine with command.



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Date: / /

Q. Write shortnote on SSH service and give examples of the clients used for the SSH service. You should also write configuration for ubuntu machine with commands.

→ SSH (Secure shell) is a network protocol that provides a secure way to access a remote computer over an unsecured network. It ensures data confidentiality & integrity by encrypting all traffic between the client and the server, protecting against eavesdropping and tampering.

Common SSH clients
- OpenSSH
- PuTTY
- Solar-PuTTY

Configuration for Ubuntu:
1. Update your package index: sudo apt update
2. Install the OpenSSH server: sudo apt install openssh-server
3. Verify the service status: sudo systemctl status ssh
4. Enable the service to start on boot: sudo systemctl enable ssh.